TaskBench: Measuring Cost-Quality Tradeoffs Across 46 LLMs and 21 Production Tasks

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Abstract

We benchmark 46 large language models from 12 API providers across 21 production tasks spanning sentiment classification, intent detection, code generation, mathematical reasoning, and 17 others. We measure both per-query cost and quality on 50 cases per task (51,580 scored cases total). Economy-tier models ($0.05–$0.49/M input tokens) show no statistically significant quality gap versus Premium models ($5+/M) on 19 of 21 tasks (Mann-Whitney $p > 0.05$; $n_{\text{Premium}} = 2$). Within providers, paying 750× more buys at most +0.13 quality points on a 1–5 scale. Per-task routing to the cheapest adequate model reduces cost by 99.7% versus the best single model with negligible quality loss. Quality scores are produced by an LLM judge validated against ground truth on three task types ($r = 0.89$–0.91). We release the full dataset, raw model responses, and evaluation prompts.[1]

1 Introduction

1.1 The Problem

Practitioners choose large language models by reputation or leaderboard rank. Existing benchmarks measure capability along a single axis: MMLU [Hendrycks et al., 2021] tests academic knowledge, Chatbot Arena [Chiang et al., 2024] aggregates human preference into a single Elo score, and HELM [Liang et al., 2023] evaluates multiple metrics but not cost. A model ranked first on Arena may cost 100× more than the fifth-ranked model for identical quality on sentiment classification. No public dataset lets a practitioner answer a concrete question: for this specific task, which model gives adequate quality at the lowest cost?

1.2 Contributions

We make three contributions.

1. **TaskBench**, a public benchmark measuring per-task cost-quality Pareto frontiers across 46 models, 21 production tasks, 12 providers, and 4 price tiers (51,580 scored cases).
2. **Empirical findings.** Economy-tier models show no significant quality gap versus Premium on 19 of 21 tasks. Per-task routing saves 99.7% versus the best single model. Within-provider price gradients are nearly flat.
3. **Judge validation.** We show that standard LLM-as-judge evaluation with generic rubrics has format bias ($r = 0.39$ versus ground truth), fix it with correctness-focused rubrics ($r = 0.91$), and release both score versions for reproducibility.

We release all raw responses, scores, and evaluation code.

[1] Data and code: `https://github.com/SebConejo/benchmark-reverse-engineering`

1.3 Key Findings Preview

Four headline results:

- *No Economy–Premium gap detected.* Mann-Whitney $p > 0.05$ on 19 of 21 tasks (the two significant results favor opposite tiers). Premium averages 4.791 versus Economy 4.741, a gap of 0.050 on the 1–5 scale.
- *Flat provider gradients.* Anthropic Haiku ($0.80/M) scores 4.775 while Opus ($15/M) scores 4.753. Google Flash ($0.15/M) matches Pro ($1.25/M). OpenAI's 750× price range yields +0.13 quality.
- *Per-task routing saves 99.7%.* The cheapest adequate model per task averages $0.000009/query versus $0.003332 for o3 alone, with at most 0.2 points quality difference on any task.
- *Five Economy models cover all tasks.* At the 4.5/5 quality threshold, five models cover all 21 tasks. The cheapest is Grok 4 Fast at $0.000435/query average.

2 Related Work

2.1 General LLM Benchmarks

HELM [Liang et al., 2023] evaluates 30 models across 42 scenarios on 7 metric families, but cost is not among them. Chatbot Arena [Chiang et al., 2024] collects over 1.5 million human preference votes, yet produces a single aggregate ranking without per-task granularity. MMLU [Hendrycks et al., 2021] and its successor MMLU-Pro test academic knowledge rather than production tasks. The Berkeley function calling benchmark covers a single task type. TaskBench adds cost as a first-class evaluation axis and provides per-task granularity across 21 diverse production tasks.

2.2 Cost-Aware LLM Evaluation and Routing

FrugalGPT [Chen et al., 2023] introduced LLM cascading for cost reduction but evaluated on one task with three models. RouterBench and RouteLLM benchmark router systems, not the underlying models. RouterArena [Lu et al., 2025] evaluates 14 routers on 8,400 queries but does not provide per-task cost-quality data for the models themselves. Artificial Analysis tracks speed and aggregate pricing but not per-task quality. We produce the underlying per-task cost-quality data that any routing system can consume.

2.3 LLM-as-Judge

MT-Bench [Zheng et al., 2023] established LLM-as-judge evaluation. Known biases include position bias, verbosity bias [Saito et al., 2023], and self-enhancement [Panickssery et al., 2024]. We identify a specific format bias that distorts cost-quality comparisons: generic rubrics reward concise formatting over correctness, systematically favoring Economy models. We quantify this ($r = 0.39 \rightarrow 0.91$) and release both V1 and V2 scores.

3 Methodology

3.1 Benchmark Design

We evaluate 21 production tasks in four categories, with 50 cases each drawn from standard datasets:

- **Classification** (4 tasks, exact-match): sentiment (SST-2), intent detection (CLINC-150, 150 classes), content moderation (ToxiGen), and multistep reasoning (ARC-Challenge).
- **Structured output** (5 tasks, LLM-judge): entity extraction, JSON transformation, named-entity recognition, function calling, and structured output generation.
- **Generation** (8 tasks, LLM-judge): code generation (HumanEval), code review, test generation, summarization (email and long-form), data-to-text, code explanation, and instruction following.
- **Reasoning and language** (4 tasks, LLM-judge): mathematical reasoning (GSM8K), extractive QA (SQuAD v2), translation EN→FR (OPUS-100), and SQL generation (Spider).

Each case has a fixed prompt sent as a single user message with no system prompt. Models receive identical inputs with temperature = 0 (or 1 for models that reject zero temperature). Two tasks have fewer than 50 cases due to dataset size: code explanation (48) and structured output (41).

3.2 Model Selection

We evaluate 46 models that completed all 21 tasks with at least 40 valid cases each, drawn from 12 API providers. We assign models to four price tiers based on input token price:

Tier	Price range (input $/M)	n
Premium	$\geq$ \$5.00	2
Standard	\$0.50–\$4.99	19
Economy	\$0.05–\$0.49	21
Micro	$<$ \$0.05	4

The two Premium models are Claude Opus 4.7 ($15/M) and GPT-5.5 Pro ($15/M). Providers include OpenAI (10 models), Qwen (7), ByteDance (5), Mistral (5), Anthropic (4), Google (4), xAI (3), Meta (3), DeepSeek (2), MiniMax (1), Moonshot (1), and Microsoft (1).

Cost is measured as per-query cost from token counts multiplied by provider-published prices at benchmark time (April–May 2026). Reasoning model costs were verified against provider billing dashboards where possible. See Section 6 for the GPT-5.5 Pro cost discrepancy.

3.3 Scoring

We use two evaluation methods:

Exact match (4 classification tasks). Model output is compared to the gold label after lowercasing and whitespace normalization. Score: 1 (correct) or 0 (incorrect), averaged to accuracy per model per task.

LLM-as-judge (17 generative tasks). GPT-4o evaluates each response on a 1–5 integer scale using task-specific correctness rubrics. Each rubric defines what scores of 1, 3, and 5 mean for that specific task. All rubrics include the anti-format instruction: "Ignore formatting, verbosity, and style. Evaluate only correctness and completeness."

Statistical methods. Bootstrap 95% confidence intervals (1,000 iterations, seed 42). Mann-Whitney U tests for tier comparisons. Kruskal-Wallis tests for multi-group comparisons.

3.4 Judge Validation

We validate the LLM judge against ground truth on three task types.

Math reasoning (GSM8K). We compare judge scores to numerical accuracy (does the model's final answer match the expected number?). With generic V1 rubrics (GPT-4o-mini judge), $r = 0.388$. With correctness-focused V2 rubrics (GPT-4o judge), $r = 0.905$. The V1 judge evaluated "reasoning quality" and penalized verbose format. Claude Opus (96% accuracy) scored 3.48/5 under V1 because of its verbose chain-of-thought. V2 evaluates whether the final answer is correct regardless of presentation.

Extractive QA (SQuAD v2). Judge scores versus exact-match accuracy. V1: $r = -0.013$ (zero correlation). V2: $r = 0.888$. The V1 judge rewarded fluent hallucinations over terse correct answers.

Code generation (HumanEval). Judge scores versus pass@1 execution rate. V2: $r = 0.891$. A naive test harness yielded only $r = 0.49$ due to indentation extraction failures; a multi-strategy harness resolved this (see Appendix C).

The V1-to-V2 improvement is driven by rubric specificity, not the judge model upgrade. V1 evaluated "quality" and rewarded concise formatting. V2 evaluates correctness and penalizes wrong answers regardless of style. The bias changed 34.7% of pairwise model rankings across all 17 LLM-judged tasks (5,195 of 14,972 pairs flipped).

We applied V2 rubrics to all 17 LLM-judged tasks (40,350 cases re-scored at approximately $60). The 14 tasks without ground truth are validated by mechanism consistency (identical rubric structure and judge model), not by direct ground-truth correlation. This is a limitation we discuss in Section 6.

4 Results

4.1 Cost-Quality Landscape

Figure 1 shows the 46×21 benchmark matrix. The dominant pattern is uniformity: most cells score above 4.5/5. Quality variance comes from 10 high-discrimination tasks where score spread exceeds 1.5 across models. On these tasks, model choice matters.

The four most discriminative tasks are intent detection (CLINC-150, spread 3.20, $\sigma = 0.624$), extractive QA (spread 2.84, $\sigma = 0.560$), content moderation (spread 2.40), and test generation (spread 2.40). The two least discriminative are structured output (spread 0.29) and multistep reasoning (spread 0.80).

For 11 of 21 tasks, the score spread is below 1.5 and model selection has minimal impact on quality. On these tasks, the practical recommendation is to use the cheapest available model.

4.2 Tier Comparison: Economy Matches Premium

Bootstrap 95% confidence intervals per tier (aggregated across all tasks):

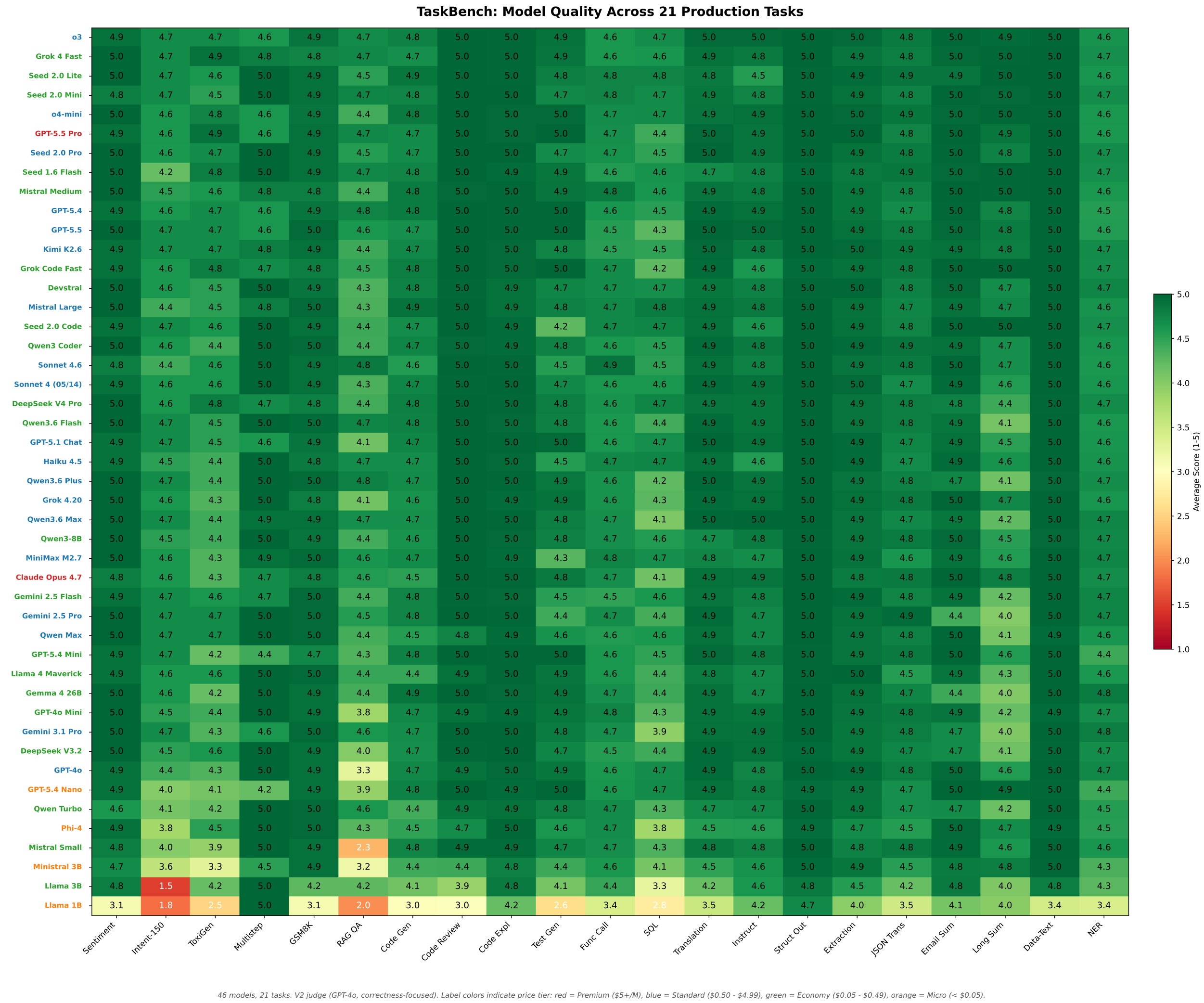


Figure 1: Quality heatmap (46 models × 21 tasks). Rows sorted by average score, columns grouped by task category. Color intensity indicates mean score per (model, task) pair on the 1–5 scale. Most cells are dark (above 4.5), with visible variation concentrated in classification and QA tasks.

Tier	Mean	95% CI	n (pairs)
Premium	4.791	[4.729, 4.853]	42
Standard	4.783	[4.760, 4.805]	399
Economy	4.741	[4.710, 4.768]	441
Micro	4.287	[4.128, 4.449]	84

The Premium-to-Economy gap is 0.050 points (1.0% of the 1–5 scale). Confidence intervals overlap heavily. Mann-Whitney tests find no significant difference on 19 of 21 individual tasks ($p > 0.05$). The two significant results go in opposite directions: Economy outperforms Premium on multistep reasoning ($p = 0.029$) while Premium outperforms Economy on instruction following ($p = 0.043$). With 21 simultaneous tests at $\alpha = 0.05$, one to two rejections are expected by chance.

Micro is the only tier with a large gap (-0.50 versus Premium). This is driven by the four Micro models, which include the two sub-3B models in our sample (Llama 1B and Ministral 3B).

Caveat. With $n = 2$ Premium models, the Mann-Whitney test has low statistical power.

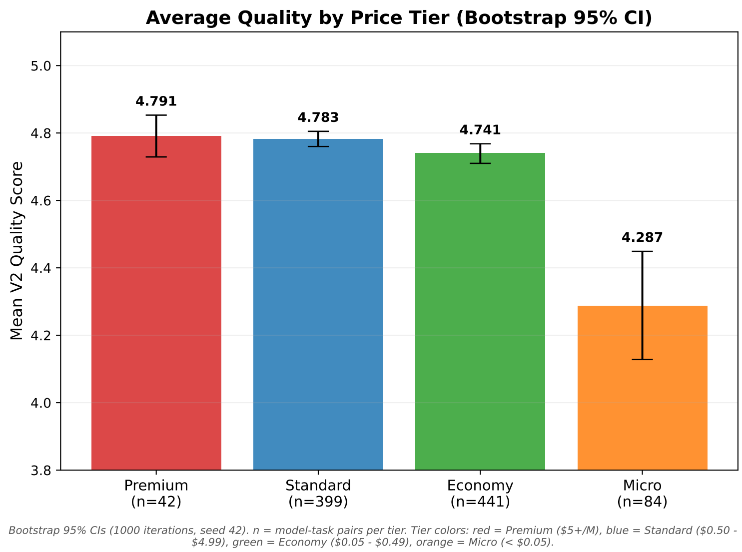


Figure 2: Average quality by price tier with bootstrap 95% confidence intervals. Premium and Economy intervals overlap on all 21 tasks. Micro is the only tier with a visible quality gap.

We report "no difference detected," not "tiers are equal." A future benchmark with five or more Premium models could detect a small gap. Even if a gap exists, it is bounded below 0.14 (the difference between the upper CI of Premium and the lower CI of Economy).

4.3 Per-Task Pareto Frontiers and Routing Savings

For each task we compute the cost-quality Pareto frontier: the set of models where no other model is both cheaper and higher quality. Premium models are almost never on the frontier. The frontier is dominated by Economy and Micro models.

Per-task routing to the cheapest model that matches o3 quality on each task saves 99.7% ($0.003332/query for o3 alone versus $0.000009/query for the routed portfolio). Even against the cheapest single model covering all 21 tasks at the 4.5 threshold (Grok 4 Fast at $0.000435/query), routing saves approximately 98%.

A portfolio of just four models covers all 21 tasks at the 4.0 quality threshold: GPT-5.4 Nano on 16 tasks, Phi-4 on 3, Qwen Turbo on 1, and Gemma 26B on 1.

4.4 Provider Gradient: Paying More Buys Little

Within each major provider, the most expensive model is not reliably the best.

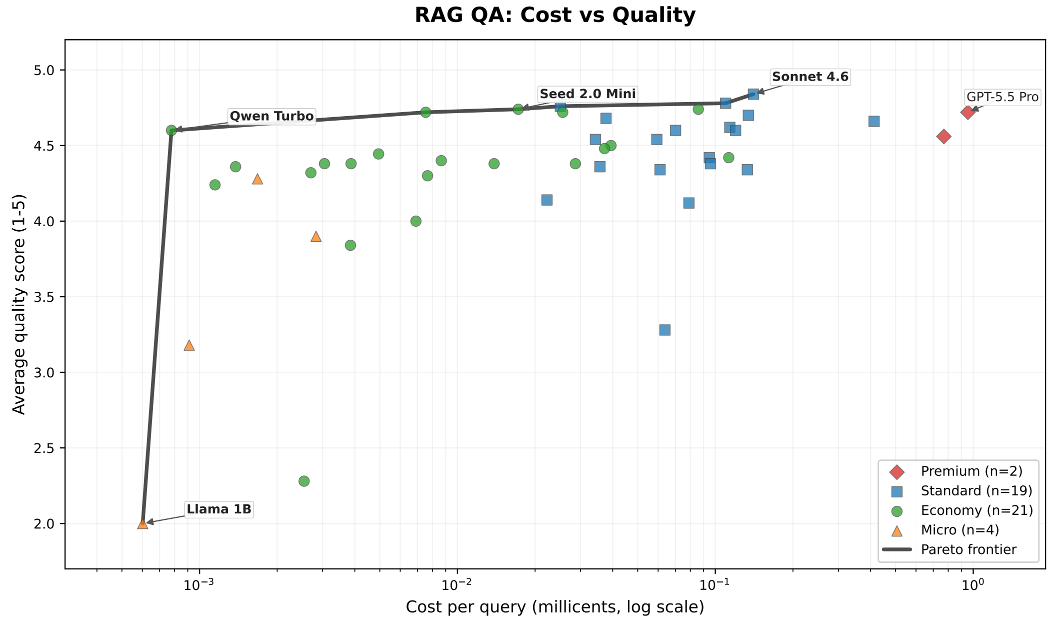
Anthropic. Haiku at $0.80/M scores 4.775. Opus at $15/M scores 4.753. Paying 19× more yields −0.022 quality.

Google. Flash at $0.15/M scores 4.749. Pro at $1.25/M scores 4.747. An 8× price increase yields −0.002.

OpenAI. Nano at $0.02/M scores 4.694. GPT-5.5 Pro at $15/M scores 4.828. A 750× price increase yields +0.134, the largest intra-provider gain in our data, and still only 2.7% of the scale.

Qwen. Turbo at $0.05/M scores 4.654. Max at $2.00/M scores 4.746. A 40× price increase yields +0.092.

Mistral. Ministral 3B at $0.04/M scores 4.441. Large at $2.00/M scores 4.800. A 50× price increase yields +0.359, the only provider where paying more substantially helps. This is driven by



n = 46 models, 50 cases each. X-axis: log scale. Black line: Pareto frontier. Tier colors: red = Premium ($5+/M), blue = Standard ($0.50 - $4.99), green = Economy ($0.05 - $0.49), orange = Micro (< $0.05).

Figure 3: Pareto frontier for extractive QA (high-discrimination task, log-scale cost axis). Sonnet 4.6 dominates at 0.14 mc; GPT-5.5 Pro costs 7× more for lower quality. Black line marks the Pareto frontier. Point shapes and colors indicate price tiers (red = Premium, blue = Standard, green = Economy, orange = Micro).

the sub-3B model floor: Ministral 3B is the smallest model in our benchmark.

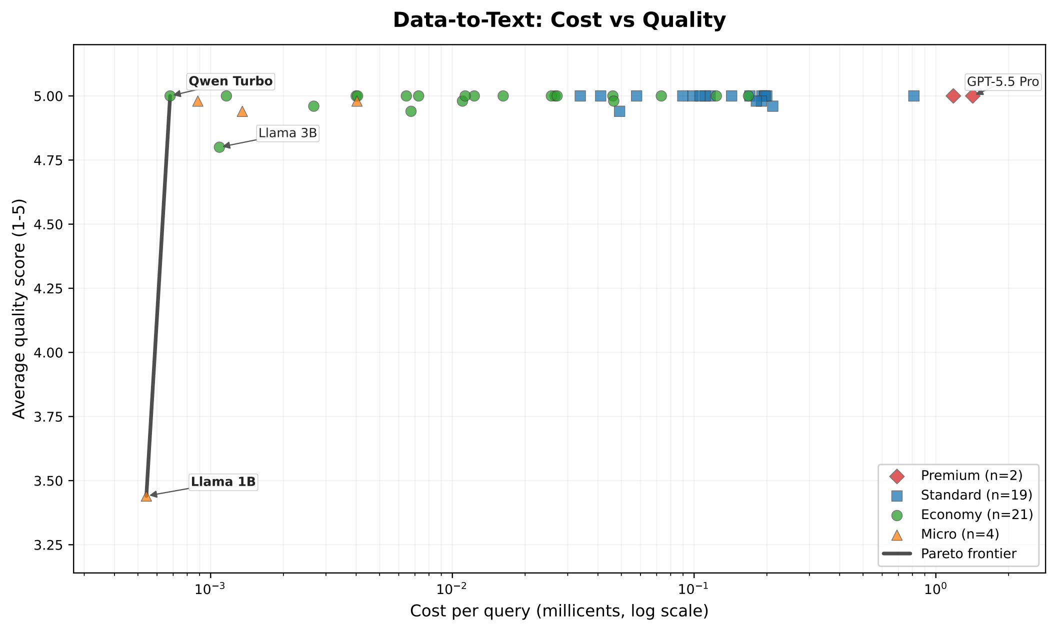
The gradient is flat at the Standard/Economy boundary. The only steep segment is from Micro to Economy, driven by sub-3B models pulling the Micro average down.

4.5 Model Selection in Practice

Given that within-provider price increases yield minimal quality returns, which models should a practitioner actually choose?

Coverage thresholds. At the 4.0/5 quality threshold, 37 of 46 models cover all 21 tasks. Model choice barely matters. At the 4.5 threshold, only five models qualify as universal: Grok 4 Fast, Seed 2.0 Mini, GPT-5.4, Seed 2.0 Lite, and o3. The cheapest is Grok 4 Fast at $0.000435/query average. At the 4.7 threshold, no single model covers all 21 tasks. The bottleneck tasks are extractive QA, intent detection, and content moderation.

Generational upgrades. The cost-quality frontier moves with each model generation. GPT-4o to GPT-5.4: −60% price ($2.50 → $1.00/M), +0.089 average quality. GPT-4o Mini to GPT-5.4 Mini: −33% price, quality flat (already near ceiling). Sonnet 4 to Sonnet 4.6: same price, +0.003 quality. The exception is DeepSeek V3.2 to V4 Pro: +211% price for +0.061 quality. In three of four provider families, the new generation is cheaper at equal or better quality. Upgrading generations is a more reliable strategy than upgrading tiers.



n = 46 models, 50 cases each. X-axis: log scale. Black line: Pareto frontier. Tier colors: red = Premium ($5+/M), blue = Standard ($0.50 - $4.99), green = Economy ($0.05 - $0.49), orange = Micro (< $0.05).

Figure 4: Pareto frontier for data-to-text (commodity task, log-scale cost axis). Qwen Turbo achieves a perfect 5.0 at 0.0007 mc; 30+ models match that score at up to 2,000× the cost. The near-flat frontier illustrates a task where model selection is cost-irrelevant.

5 Discussion

5.1 Practical Implications

Three recommendations for practitioners:

1. **Default to Economy tier.** On 19 of 21 tasks, Economy models showed no significant quality difference versus Premium (the two exceptions favor opposite tiers). The safest default is the cheapest model scoring at least 4.5 on the target task.

2. **Route per task for maximum savings.** A four-model portfolio covers all 21 tasks at the 4.0 threshold and costs $0.000009/query average. The savings come from commodity tasks where Micro-tier models suffice.

3. **Upgrade generations, not tiers.** GPT-5.4 is 60% cheaper than GPT-4o at higher quality. The cost-quality frontier moves down-left with each generation. Paying for a premium tier within the same generation has near-zero return.

Premium models may add marginal value on open-ended language tasks, but the effect size (< 0.2 points) is within our task-level confidence intervals.

5.2 Evaluation Methodology Implications

The format bias finding (Section 3.4) generalizes beyond this benchmark. Any LLM-as-judge evaluation using generic "rate quality 1–5" rubrics likely contains this bias. We recommend three practices: (1) always validate against ground truth on at least two task types, (2) use task-specific

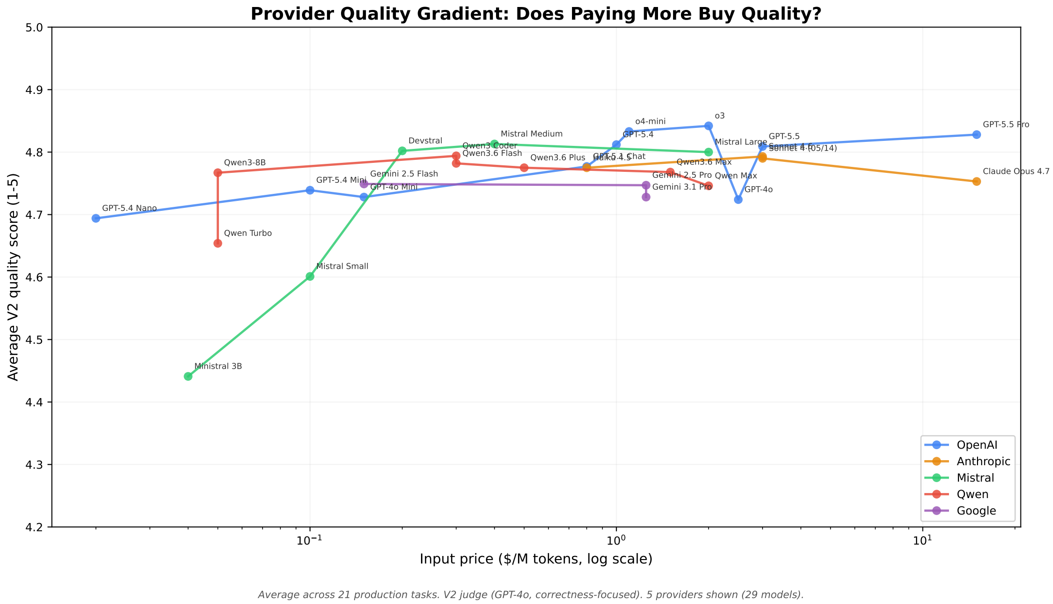


Figure 5: Within-provider price-quality gradient. Each line connects models from the same provider, ordered by input price. The gradient is nearly flat for Anthropic, Google, and Qwen. OpenAI shows the largest absolute gain (+0.13) across its 750× price range. Mistral is the exception, driven by its sub-3B entry-level model.

correctness anchors rather than generic quality rubrics, and (3) report both judge scores and native metrics where available.

Verbosity is not the primary driver of the bias ($r = 0.22$, $p = 0.13$ between verbosity ratio and V1-to-V2 score delta). The mechanism is rubric ambiguity: generic rubrics allow the judge to reward format features that correlate with model price tier.

6 Limitations

- $n = 2$ **Premium models.** The Economy-matches-Premium finding is "no difference detected," not "tiers are equal." Two Premium models (Claude Opus 4.7 and GPT-5.5 Pro) provide low statistical power. Future work should include five or more Premium models.
- **Judge validated on 3 of 17 LLM-judged tasks.** The remaining 14 tasks are validated by mechanism consistency (identical rubric structure and GPT-4o judge), not by direct ground-truth correlation.
- **50 cases per task.** Confidence intervals are approximately ±0.2. Differences below 0.3 are not distinguishable from noise. This is standard for benchmark papers; HELM uses similar per-scenario sample sizes.
- **Single-turn only.** Multi-turn conversations, vision, and audio inputs are out of scope.
- **Price snapshot (April–May 2026).** LLM prices change monthly. Tier-level findings are more durable than absolute prices because relative pricing between tiers rarely inverts. We release raw data for re-computation with current prices.

- **Potential data contamination.** Models may have seen GSM8K, SQuAD, and HumanEval during training. This affects absolute scores but not relative comparisons, since all models face the same contamination risk on standard datasets.
- **Reasoning model cost tracking.** GPT-5.5 Pro and o3 reasoning tokens are partially invisible in the API response. Tracked cost ($143.81 total spend) underestimates real spend (approximately $180+) for these models specifically. Non-reasoning model costs remain accurate.
- **Model diversity.** Our sample includes three geographic origins and both open-weight and closed-source licenses. Neither dimension significantly predicts quality: provider origin shows no effect (Kruskal-Wallis $H = 1.83$, $p = 0.40$), and the open-versus-closed gap (0.183, $p = 0.0004$) is driven by sub-3B open-weight models (see Appendix G).
- **English-only** (except the EN→FR translation task).

7 Conclusion

TaskBench provides a systematic per-task cost-quality benchmark for production LLM use. Across 46 models and 21 tasks, we find that the cost-quality curve is flat: on 19 of 21 tasks we detect no significant difference between Economy and Premium tiers, and the two significant results favor opposite tiers ($n_{\text{Premium}} = 2$; more Premium models are needed to confirm this pattern). The practical implication is that practitioners should default to Economy models and use per-task routing for maximum savings.

We also show that standard LLM-as-judge evaluation has a format bias that specifically distorts cost-quality comparisons. Correctness-focused rubrics fix this with minimal effort.

We release all 51,580 scored cases, 51,705 raw model responses, evaluation prompts, and analysis code. We encourage the community to extend TaskBench with more Premium models, multi-turn tasks, and longitudinal price tracking.

Acknowledgments and Disclosure

Disclosure. This work was conducted by Sébastien Conejo, co-founder of Manifest (https://github.com/mnfst/manifest), an open-source LLM model router. Manifest was not used in this benchmark. The routing savings analysis (Section 4.3) applies to any routing system and is computed from the benchmark data using a simple per-task argmin. All code, data, and evaluation prompts are publicly available.[2]

References

Lingjiao Chen, Matei Zaharia, and James Zou. FrugalGPT: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023.

Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, et al. Chatbot arena: An open platform for evaluating LLMs by human preference. *arXiv preprint arXiv:2403.04132*, 2024.

[2] https://github.com/SebConejo/benchmark-reverse-engineering

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *Proceedings of the International Conference on Learning Representations (ICLR)*, 2021.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *Transactions on Machine Learning Research (TMLR)*, 2023.

Yifan Lu, Rixin Liu, Jiayi Yuan, Xingqi Cui, Shenrun Zhang, Hongyi Liu, and Jiarong Xing. RouterArena: An open platform for comprehensive comparison of LLM routers. *arXiv preprint arXiv:2510.00202*, 2025.

Arjun Panickssery, Samuel R. Bowman, and Shi Feng. LLM evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*, 2024.

Keita Saito, Akifumi Wachi, Koki Wataoka, and Youhei Akimoto. Verbosity bias in preference labeling by large language models. *arXiv preprint arXiv:2310.10076*, 2023.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, et al. Judging LLM-as-a-judge with MT-Bench and chatbot arena. In *Advances in Neural Information Processing Systems*, 2023.

A Full Pareto Frontiers (21 Tasks)

One single-panel log-scale Pareto figure per task, following the style of Figures 3 and 4. Figures are available in `results/figures/` in the companion repository.

B Complete Model Table

Table with all 46 complete models: provider, model name, price tier, input price/M, output price/M, average score, rank, and number of valid cases. Available as `task_model_aggregates.json` in the companion repository.

C Judge Validation Details

V1 versus V2 correlation summaries for all four validated task types:

#	**Task**	**V1** r	**V2** r	**Native metric**
1	Math reasoning (GSM8K)	0.388	0.905	Numerical accuracy
2	Extractive QA (SQuAD)	−0.013	0.888	Substring match
3	Code generation (HumanEval)	0.491*	0.891	pass@1
4	Translation (OPUS-100)	n/a	0.290	BLEU (single ref.)

*V1 code correlation uses a naive harness; V2 uses the multi-strategy harness.

The code generation validation required a multi-strategy test harness. A naive harness using `textwrap.indent` failed on 47.2% of responses due to indentation mismatches. Our multi-strategy approach tries four extraction strategies in sequence (standalone function extraction, uniform indent, selective indent, raw append), resolving 91.4% of formatting failures and raising the pass rate from 44.9% to 91.4%.

Translation BLEU is inconclusive ($r = 0.29$) due to the metric's single-reference limitation, not the judge's quality.

Per-task ranking flip rates range from 20.2% (code review, Kendall $\tau = 0.596$) to 50.3% (extractive QA, $\tau = -0.006$). The aggregate flip rate is 34.7% (5,195 of 14,972 pairs).

D Judge Prompts

Full V1 and V2 judge prompt text for all 17 LLM-judged tasks. Available in `JUDGE_PROMPTS_V2.md` in the companion repository.

E Statistical Tables

Bootstrap 95% confidence intervals per (task, tier) and Mann-Whitney U p-values per task (Economy versus Premium, Economy versus Standard). Coverage matrix showing per-model threshold coverage at 4.0, 4.5, and 4.7. Per-task routing savings (cheapest adequate model, cost, and savings percentage). All data available in `results/analysis_v3_final/` in the companion repository.

F Top-5 Models Per Task

For each of the 21 tasks, the five highest-scoring models ranked by quality (ties broken by lowest cost). Available as `top5_per_task.json` in the companion repository.

G Model Diversity Analysis

Provider origin. Three geographic groups: Chinese ($n = 16$, mean 4.779), American ($n = 26$, mean 4.682), and European ($n = 5$, mean 4.691). Kruskal-Wallis $H = 1.83$, $p = 0.40$. No significant difference. The American group has the widest range (3.394 to 4.842), driven by sub-3B Llama models at the low end.

License type. Open-weight ($n = 18$, mean 4.603) versus closed-source ($n = 30$, mean 4.786). Mann-Whitney $p = 0.0004$, gap 0.183. However, the open-weight group includes all sub-3B models (Llama 1B, Llama 3B, Ministral 3B). Excluding sub-3B models narrows the gap to approximately 0.066. The license effect is driven primarily by model size, not by the open-weight development process.

H Task Discriminativeness

Horizontal bar chart showing score spread (maximum minus minimum) per task across all 46 models, with tasks color-coded by category. Available as `task_discriminativeness_v3.svg` in the companion repository.